Provenance Tracing in Network Diffusion Algorithms

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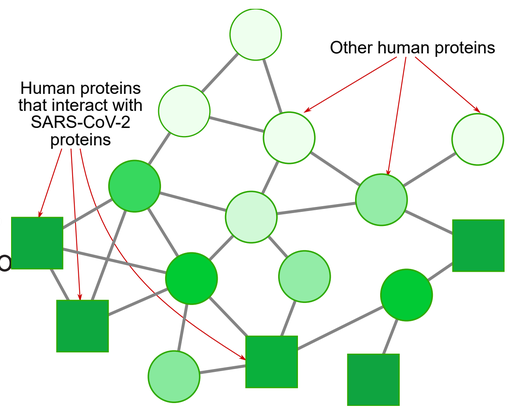
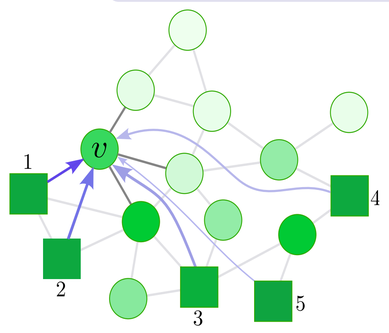
Framework of Network Diffusion

- Input: a set of experimentally determined nodes (seed nodes) having an association with a certain phenotype, e.g., human SARS-CoV-2 interactors[1].
- Approach: Propagate score from seed nodes to other nodes, taking network structure into account and ensuring smoothness of scores.
- Output: Predictions, i.e., nodes with high scores.

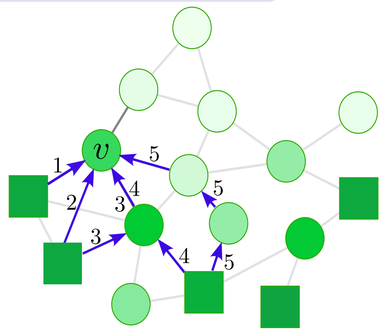


Provenance Tracing

Trace the origin of a prediction back to seed nodes.



- **Node-based:** Rank each seed node based on its 'contribution' to a node's score.



- **Path-based:** Rank each path starting at a seed node based on its 'contribution' to a node's score.

Motivation

- Provenance tracing provides interpretation for scores computed by network diffusion algorithms.
- Lack of interpretation leads to unreliable predictions causing significant expenditure of time and financial resources if pursued in downstream tasks, e.g., drug repurposing[2].

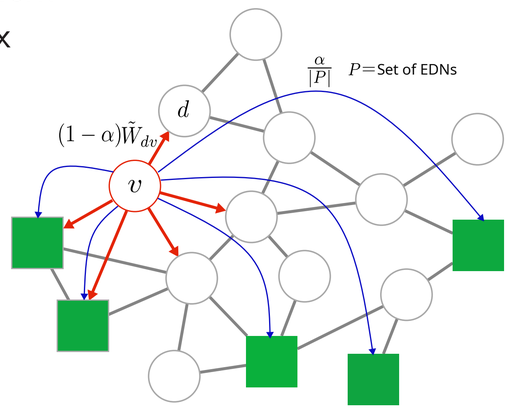
Random Walk with Restart (RWR)

- $G = (V, E, w)$: undirected, weighted network
- $\tilde{W}$: Normalized weighted adjacency matrix
- $\frac{\alpha}{|P|}\vec{y}$: Restart probability vector
- $\vec{s}$: Predicted score vector

$$s_v = \sum_{u \in P} K_{vu},$$

where $K = \frac{\alpha}{|P|}(I - (1-\alpha)\tilde{W})^{-1}$

- s_v is a linear sum of the *contributions* K_{vu} from every seed node u.

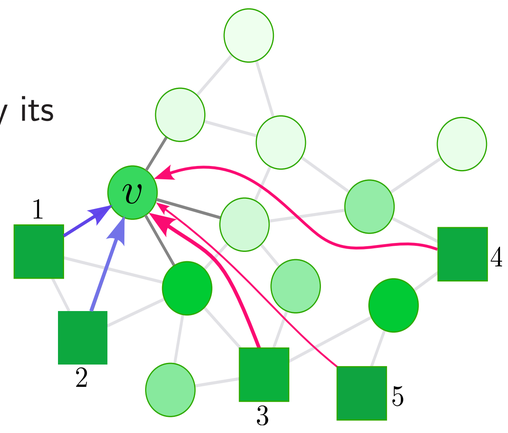


Node-based Effective Diffusion (NED)

The fraction of a node v's score contributed by its non-neighboring seed nodes.[3]

$$NED(v) = 1 - \frac{\sum_{u \in P \cap N(v)} K_{vu}}{s_v},$$

where $N(v)$ = Set of neighbors of node v

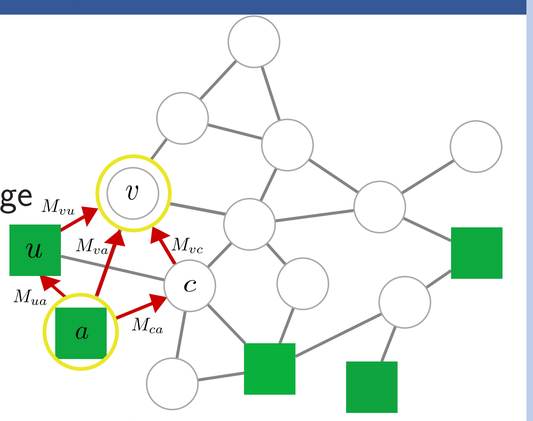


Path-based Effective Diffusion (PED)

Define M such that

- M_{vu}: Contribution of node u to s_v via edge (u, v)
- Contribution from a path is a product of edge weights along the path

s_v = Weighted sum of contribution from all u–v paths, where u ranges over seed nodes.



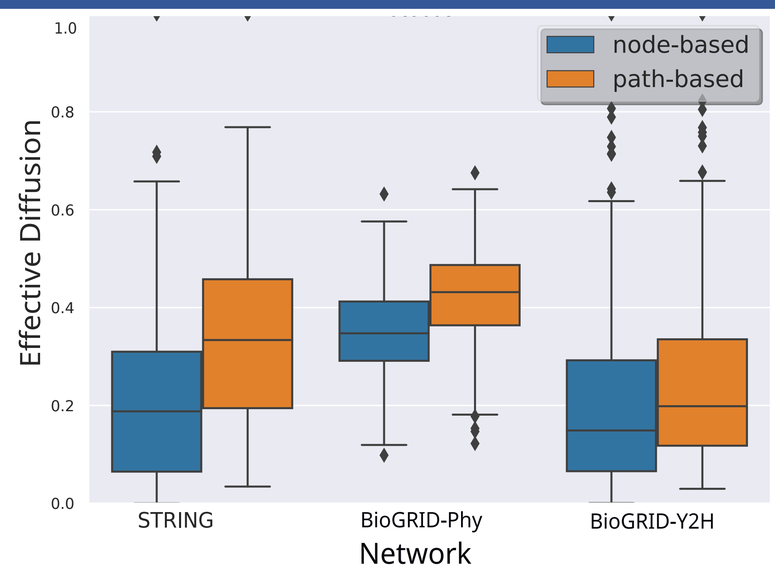
PED is the fraction of a node's score arising via paths of length > 1.

$$PED(v) = 1 - \frac{\frac{\alpha}{|P|}\sum_{u \in P} M_{vu}}{s_v}$$

References

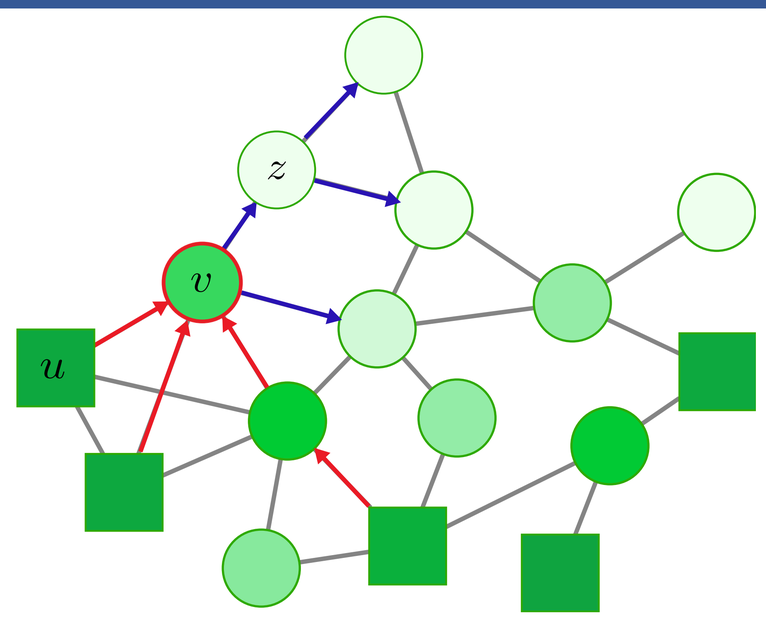
1. Gordon *et al.* A SARS-CoV-2 human protein-protein interaction map reveals [...] *Nature* 2020
2. Kasif and Roberts. We need to keep a reproducible trace of facts, predictions [...] *PLoS Biology* 2020
3. Law *et al.* Interpretable network propagation with application to expanding [...] *GigaScience* 2021

Node-based vs. Path-based Effective Diffusion



Path-based effective diffusion is always significantly higher than node-based (Mann-Whitney U test p-value $< 8.08 \times 10^{-10}$)

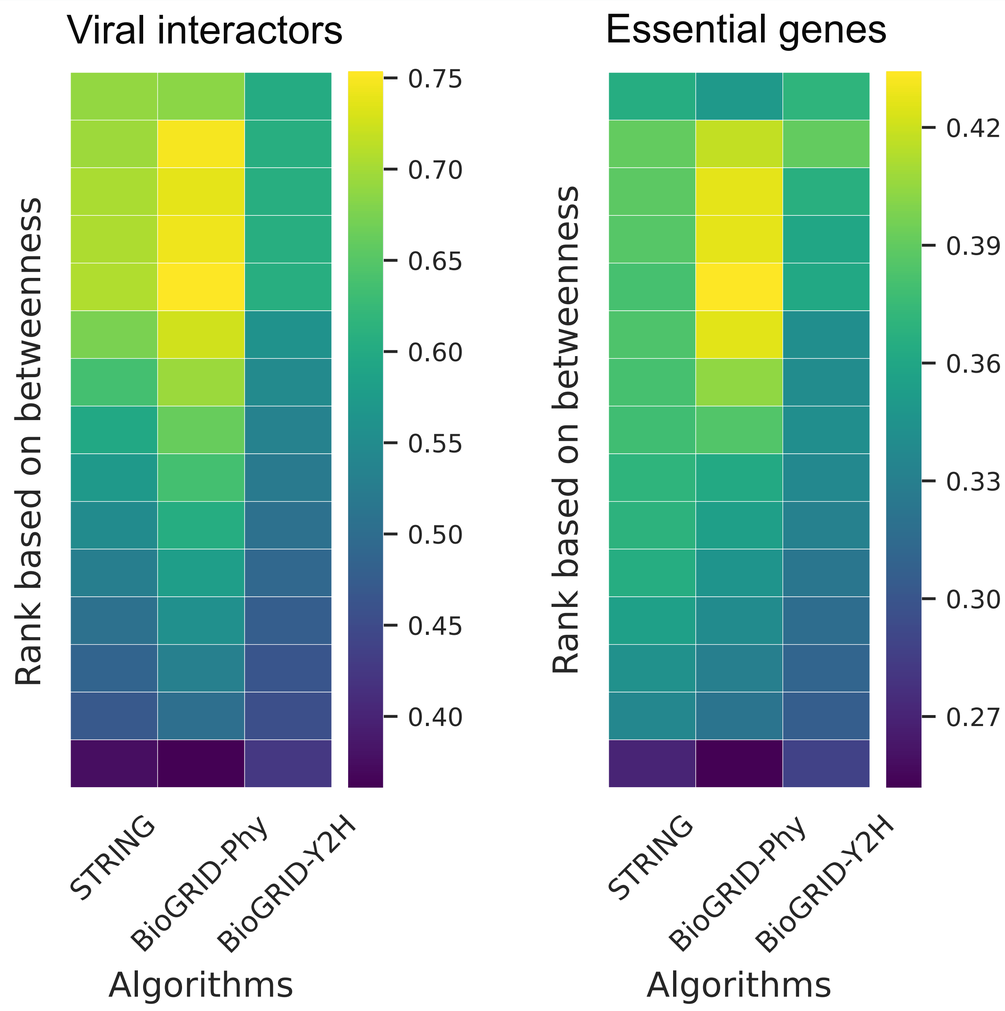
Diffusion Betweeness



For a node v, the sum of the contribution made to any node j's score along all π_{i-v-j} paths for which starting node i is any seed node and v is an internal node.

$$b(v) = \frac{\alpha}{|P|} \sum_{i \in P,\, j \in V} \text{contribution}(\pi_{i-v-j})$$

Biological Relevance of Diffusion Betweenness



- Proteins with high diffusion betweenness are enriched with viral interactors (Fisher's exact test p-value < 0.01).
- Proteins with high diffusion betweenness are enriched with essential genes (Fisher's exact test p-value < 0.01).

Summary

- Introduced path-based method to trace provenance of predictions made by random-walk-based diffusion algorithms.
- Elucidates how the non-local structure of the network is exploited to propagate scores from seed nodes to other nodes.
- Diffusion algorithms exploit biologically significant nodes in propagating score.
- Research is a step forward in the field of interpretation with the ultimate goal of establishing the reliability of computational models in high-stake tasks.

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